LAB 18

Diagnostics Stage: Automated Bias Checks before Go-Live

Item	Detail
Topic	Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis
Pipeline stage	Diagnostics
TSC reference	A9, K8
Duration	20 min
Folder	labs_fullcode/Lab18-diagnose-bias/
Files	Lab18-diagnose-bias/diagnose_stage.py, Lab18-diagnose-bias/buggy_backtest.py, pipeline/diagnose.py, Lab18-diagnose-bias/solution/diagnose.py
Prompts	Lab18_Prompts.pdf in this folder (step -> prompt, ready to copy)
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

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Scenario

A junior quant sends the investment committee a 'smoothed momentum' backtest on SPY with a Sharpe ratio above 4 and asks to go live. The desk responds by adding a diagnostics stage to the pipeline: every strategy - including its own ema_cross - must pass automated checks for look-ahead, cost drag, overfitting and survivorship before it may trade, even on paper.

Objective

Build `python -m pipeline diagnose` to run automated look-ahead (features and execution), cost-drag, overfitting and survivorship checks on any strategy and on the junior quant's exhibit, and write a PASS / WARN / FAIL report with a go-live verdict.

Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

How to Run the Python Scripts

Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.

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Step	Command / file
Open a terminal in the labs_fullcode folder	cd labs_fullcode
Create and activate a virtual environment (once)	python3 -m venv .venv source .venv/bin/activate (Windows: .venv\Scripts\activate)
Install the libraries (once)	pip install -r requirements.txt
Optional: your Alpaca paper keys (leave blank to run OFFLINE)	cp .env.example .env (Windows: copy .env.example .env) then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env
Run Lab 18	python3 Lab18-diagnose-bias/diagnose_stage.py (same as: python -m pipeline diagnose)
Also in this lab: buggy_backtest.py	python3 Lab18-diagnose-bias/buggy_backtest.py
The complete code	pipeline/diagnose.py (reference copy: Lab18-diagnose-bias/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

Lab Steps - In Python with Your AI Assistant

AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab18-diagnose-bias/Lab18_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.

Step 1: Run the suspicious backtest (A9)

Run: python3 Lab18-diagnose-bias/buggy_backtest.py. Record the CAGR, Sharpe ratio and win rate.

Check: CAGR 71.52%, Sharpe 4.28, win rate 67.5% - far above anything a real SPY timing rule achieves.

Step 2: Find the problems (A9, K8)

Read the three lines marked CHECK 1 to 3 and write down what is wrong with each.

AI prompt - paste into your AI coding assistant
Review labs_fullcode/Lab18-diagnose-bias/buggy_backtest.py as a model-risk reviewer. For each line marked CHECK 1 to 3, explain the bias, how it inflates the result and the correct fix. Do not run or change the file.

Check: CHECK 1 centred window (uses t+1 and t+2), CHECK 2 same-bar returns, CHECK 3 no costs.

Step 3: Build the look-ahead checks (A9, K8)

Ask your assistant for the first part of pipeline/diagnose.py.

AI prompt - paste into your AI coding assistant
Create labs_fullcode/pipeline/diagnose.py. buggy_signal(bars) reproduces the exhibit (5-day rolling mean with center=True, long when it rises); trailing_signal(bars) is the fix (trailing window). In run(strategy=None, trials=None) add check 1, look-ahead in FEATURES, using pipeline.signals.lookahead_test (25 truncation truncation and future-shock checks) on the configured strategy and on the exhibit (fn=buggy_signal); and check 2, look-ahead in EXECUTION: the strategy's Sharpe with same-bar returns (target_t x return_t) versus the pipeline's next-bar backtest, plus the exhibit fixed one problem at a time: 0 as submitted (centred, same bar), 1 trailing window same bar, 2 + next-bar execution (common.backtest.backtest), 3 + alpaca_retail costs, and SPY buy and hold, as a CAGR / Sharpe / max drawdown table. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: The exhibit fails the truncation test; the configured strategy passes.

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Step 4: Build the cost, overfitting and survivorship checks (A9, K8)

Ask your assistant for checks 3-5, the report and main().

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/diagnose.py add: check 3 cost drag - gross vs net CAGR from cost_summary(), break-even from pipeline.backtest.break_even_bps versus the assumed one-way cost (half spread + slippage of the config preset): PASS if break-even >= 3 x assumed and drag < 25% of gross; check 4 overfitting - random_strategies(spy_returns, 500, seed=1) (skill-free long/flat signals switching with 10% probability a day, traded next bar), the best in-sample Sharpe among the first N where N = the number of sets in _outputs/Lab10_grid_results.csv (default 76), compared with the strategy's 2018-2021 Sharpe, and its 2022-2025 decay; check 5 survivorship / hindsight - the universe was chosen from today's listed symbols, and an equal-weight NVDA/MSFT/AAPL 'AI basket' versus SPY. Print each check as [PASS|WARN|FAIL|INFO] check subject evidence, a VERDICT ('BLOCK - fix the FAIL items' if the strategy has a FAIL, else 'CLEAR for paper trading - review every WARN'), write _outputs/Lab18_diagnostics.csv and charts Lab18_bias_comparison.png and Lab18_overfitting.png. main(argv): --strategy, --trials, --strict (exit 1 on BLOCK). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline diagnose --help shows --strategy, --trials and --strict.

Step 5: Run the diagnostics (A9)

Run: python -m pipeline diagnose (same as: python3 Lab18-diagnose-bias/diagnose_stage.py).

Check: ema_cross: look-ahead PASS (0 of 25), execution PASS (same-bar Sharpe 1.45 vs next-bar 0.82), cost drag PASS (gross 9.13% -> net 9.02%, break-even 135.8 bps vs 1.5 bps), overfitting WARN (IS Sharpe 0.86 vs best of 76 random 1.42; OOS 0.75), survivorship WARN. Exhibit: look-ahead FAIL (25 of 25), execution FAIL (4.28 -> 0.24). VERDICT for ema_cross: CLEAR for paper trading - review every WARN.

Step 6: Read the exhibit fixed one problem at a time (A9, K8)

Read the fix table and open _outputs/Lab18_bias_comparison.png.

Check: As submitted 71.47% / Sharpe 4.28; trailing window, same bar 66.56% / 4.68; + next-bar execution 2.23% / 0.24; + realistic costs 1.46% / 0.18; SPY buy and hold 14.14% / 0.78 - the same-bar error is the main bias.

Step 7: Read the overfitting and survivorship evidence (K8)

Read the INFO row and open _outputs/Lab18_overfitting.png. Then run python -m pipeline diagnose --strategy dip_buy.

Check: 500 random strategies: best IS Sharpe 1.77 -> OOS 0.55, median OOS 0.46, IS/OOS correlation -0.05, 16% have IS Sharpe > 1. AI basket CAGR 38.84% vs SPY 14.15%. dip_buy: cost drag PASS (break-even 19.5 bps), overfitting WARN (IS Sharpe 0.14, OOS 0.80).

Step 8: Use the stage as a gate and write the committee note (A9)

Run: python -m pipeline diagnose --strict and check the exit code (0 for ema_cross). The e2e capstone uses the same verdict to block orders.

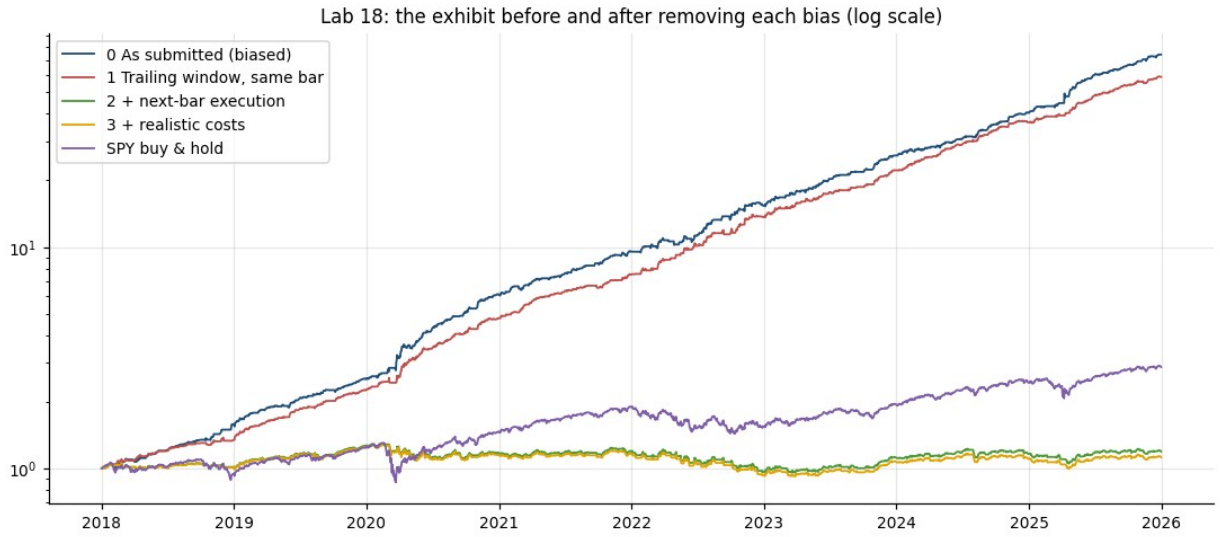
AI prompt - paste into your AI coding assistant
Using labs_fullcode/_outputs/Lab18_diagnostics.csv, write a 4-sentence note for the investment committee: why the junior quant's Sharpe of 4.28 is not credible, which automated check caught each problem, what the configured ema_cross strategy's WARN items mean, and what must happen before any live capital.

Check: Exit code 0 for ema_cross; a note that names look-ahead (features and execution), cost drag, overfitting and survivorship.

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Expected Output (Full Code)

python -m pipeline diagnose: ema_cross PASS for look-ahead (0 of 25), execution (same-bar 1.45 vs next-bar 0.82) and cost drag (break-even 135.8 bps vs 1.5 bps), WARN for overfitting (IS 0.86 vs best of 76 random 1.42) and survivorship -> CLEAR for paper trading; exhibit FAIL: 25 of 25 look-ahead checks, Sharpe 4.28 as submitted -> 0.24 fixed -> 0.18 with costs; 500 random strategies best IS 1.77 -> OOS 0.55; AI basket 38.84% vs SPY 14.15%; files _outputs/Lab18_diagnostics.csv and charts _outputs/Lab18_bias_comparison.png and _outputs/Lab18_overfitting.png.



Lab 18 output (labs_fullcode/_outputs/Lab18_bias_comparison.png), offline mode on the bundled data.

Debrief

Most 'too good to be true' results come from a short list of causes (K8): look-ahead in features or execution, overfitting, survivorship and ignored costs. Automating the checks and using the verdict as a gate (A9) is what stops an unintended outcome from reaching even a paper account - and the same checks run on the desk's own strategy, not only on other people's.

Troubleshooting

Symptom	What to do
Step 1 shows a slightly different CAGR from the fix table	buggy_backtest.py drops the first NaN rows; the stage fills them with 0 - 71.52% vs 71.47%.
Random-strategy numbers differ	Change the seed or --trials and the threshold changes, but the pattern (no persistence) stays.
Is center=True really look-ahead?	Yes: a centred 5-day mean at day t uses the closes of t+1 and t+2.

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